→Name and purpose of the application:-

Application Name:-

Airline Reservation & Flight Booking System

Purpose:-

To manage flight schedules, aircraft inventory, passenger profiles, ticket bookings, seat assignments, and payment records for an airline service provider.

→List of all required tables

1.Passengers

2.Airports

3.Flights

4.Bookings

5.Ticket_Details

6.Payments

→Columns/attributes for each table:-

Passengers:-

Passenger id (Primary Key, INT, Auto Increment)

first name (VARCHAR(50), NOT NULL)

last name (VARCHAR(50), NOT NULL)

email (VARCHAR(100), UNIQUE, NOT NULL)

phone number (VARCHAR(15), NOT NULL)

Passport number (VARCHAR(20), UNIQUE, NOT NULL)

nationality (VARCHAR(50))

Airports:-

Airport code (Primary Key, CHAR(3), e.g., 'DEL', 'JFK')

Airport name (VARCHAR(100), NOT NULL)

city (VARCHAR(50), NOT NULL)

country (VARCHAR(50), NOT NULL)

Flights:-

Flight id (Primary Key, INT, Auto Increment)

Flight number (VARCHAR(10), NOT NULL)

Origin airport code (Foreign Key referencing Airports(airport_code))

Destination airport code (Foreign Key referencing Airports(airport code))

Departure time (DATETIME, NOT NULL)

Arrival time (DATETIME, NOT NULL)

Total seats (INT, NOT NULL)

Base price (DECIMAL(10, 2), NOT NULL)

Bookings

Booking id (Primary Key, INT, Auto Increment)

Passenger id (Foreign Key referencing Passengers(passenger id))

Booking reference (VARCHAR(8), UNIQUE, NOT NULL, e.g., PNR code)

Booking date (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

Total fare (DECIMAL(10, 2), NOT NULL)

status (ENUM('Confirmed', 'Pending', 'Cancelled'), DEFAULT 'Pending')

Ticket Details:-

Ticket id (Primary Key, INT, Auto Increment)

Booking id (Foreign Key referencing Bookings(booking_id))

Flight id (Foreign Key referencing Flights(flight_id))

Seat number (VARCHAR(5), NOT NULL)

Travel class (ENUM('Economy', 'Premium Economy', 'Business', 'First Class'), NOT NULL)

fare (DECIMAL(10, 2), NOT NULL)

Payments:-

Payment id (Primary Key, INT, Auto Increment)

Booking id (Foreign Key referencing Bookings(booking id))

Payment method (ENUM('Credit Card', 'Debit Card', 'Net Banking', 'UPI'))

Payment status (ENUM('Pending', 'Success', 'Failed', 'Refunded'), DEFAULT 'Pending')

Transaction date (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

→Relationships Between Tables

Parent Table	Parent Key	Child Table	Foreign Key	Cardinality	Relationship Description
Passengers	passenger_id	Bookings	passenger_id	1 to Many (1:N)	A passenger can create multiple booking records over time.
Airports	airport_code	Flights	origin_airport_code	1 to Many (1:N)	One departure airport hosts multiple outgoing flights.
Airports	airport_code	Flights	destination_airport_code	1 to Many (1:N)	One arrival airport

Parent Table	Parent Key	Child Table	Foreign Key	Cardinality	Relationship Description
					receives multiple incoming flights.
Bookings	booking_id	Ticket_Details	booking_id	1 to Many (1:N)	One booking reservation can contain multiple flight ticket segments (e.g., multi-leg or round trips).
Flights	flight_id	Ticket_Details	flight_id	1 to Many (1:N)	One flight instance has

Parent Table	Parent Key	Child Table	Foreign Key	Cardinality	Relationship Description
					multiple individual tickets/ seats sold.
Bookings	booking_id	Payments	booking_id	1 to 1 (1:1)	Each overall booking reservation has one corresponding payment transaction.

→Purpose of Each Table:-

Passengers:

Stores traveller profile details, contact information, and government identity metrics (passport) necessary for international and domestic flight manifests.

Airports:

Maintains a lookup directory of world airports, complete with standard 3-letter IATA codes, cities, and country locations.

Flights:

Tracks flight schedules, origin and destination pairings, capacity limits, and flight departure/arrival timings.

Bookings:

Stores master reservation header details, including unique PNR codes, overall cost, and reservation statuses linked to the primary passenger.

Ticket Details:

Resolves multi-flight itineraries by storing specific ticket segments, individual assigned seats, and cabin class selections tied to a main booking.

Payments:

Manages monetary transactions, payment mode logs, and verification statuses associated with flight reservations.